

Assignment: Training Neural Networks

DSA 8401: Applied Machine Learning

Master in Data Science and Analytics



Strathmore University

@iLabAfrica Centre

1 Goals

This 3 weeks' activity focuses on learning the basic procedure to train fully connected artificial neural networks. The students must learn:

- Implement fully connected Neural Networks
- Train and evaluate the model
- Fine tune hyperparameters
- Make predictions with the model
- Saving and restoring models

2 Assignment Description

In this activity, we will use a dataset with medical records to train a neural network to predict when a patient has diabetes. The description of the dataset can be found at the following link:

<https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database>.

It is requested:

1. Use Keras to load the dataset
2. Create a model using the sequential API trying different number of layers (1-5) and different number of neurons per layer.
3. Train the different models and evaluate. Analyze which one perform better.
4. Fine tune learning rate and batch size.
5. Save the best model and hyperparameters.
6. Compare the performance of the Adam optimizer with momentum and another optimizer you choose.
7. Try regularization and dropout to avoid overfitting. Analyse the results

3 Hand in

The students will upload in the eLearning platform the Jupyter notebook. The notebook will contain the python code, the plots and especially personal responses to the previous questions. If there is no personal comment the mark will be F. Marks C,B and A will depend on the variety and quality of the personal comments.